IGCSE 0607 Practice Questions — E10.1

Classifying statistical data

Total Questions: 9 — **Total Marks:** 60 Marks

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1 70 students are asked to choose their favourite sport from Football, Tennis, or Basketball.
The incomplete two-way table shows some information about their choices.

(a) Complete the two-way table.

	Football	**Tennis**	**Basketball**	**Total**
Boys	15		12	
Girls		14		30
Total	23		20	70

[3]

(b) A student is chosen at random.
Find the probability that the student is a boy whose favourite sport is Tennis.

..... **[1]**

2 50 students each choose one sport to play: Tennis, Football, or Swimming.
The students are classified as Boys or Girls.

(a) Complete the two-way table to show this information.

	Tennis	**Football**	**Swimming**	**Total**
Boys	7			28
Girls	8		6	
Total		20		50

[3]

(b) A student is chosen at random.
Find the probability that the student is a boy who chose Football.

..... **[1]**

3 80 students were asked to choose their favourite sport from Football, Basketball, and Tennis.

- There are 35 girls in total.
- 38 students chose Football, of which 24 were boys.
- 12 girls chose Basketball.
- 20 students chose Tennis.

(a) Complete the two-way table.

	Football	**Basketball**	**Tennis**	**Total**
Boys				
Girls				
Total				

[3]

(b) A student is chosen at random.
Find the probability that the student is a boy whose favourite sport is Basketball.

..... **[1]**

(c) A girl is chosen at random.
Find the probability that her favourite sport is Tennis.

..... **[2]**

4 A survey of adults recorded their preferred hot drink, either Tea or Coffee.
Let x be the number of males who prefer Coffee.
The number of males who prefer Tea is $2x + 10$.
The number of females who prefer Coffee is $x + 5$.
The number of females who prefer Tea is $3x - 11$.

(a) Write down an expression, in terms of x, for the total number of adults.
Give your answer in its simplest form.

..... **[2]**

(b) The survey included 130 adults.
Find the value of x.

$x =$ ..... **[2]**

(c) Complete the two-way table.

	Coffee	**Tea**	**Total**
Male			
Female			
Total			

[3]

(d) One of the adults who prefers Coffee is chosen at random.
Find the probability that this adult is female.

..... **[2]**

5 A group of 120 adults were asked whether they prefer Tea or Coffee.

- 55 of the adults are women.
- 70 of the adults prefer Tea.
- 25 of the men prefer Coffee.

(a) Complete the two-way table to show this information.

	Tea	**Coffee**	**Total**
Men			
Women			
Total			

[3]

(b) Work out the number of women who prefer Tea as a percentage of the total number of adults.

..... % **[2]**

6 In a school of 400 students, the ratio of juniors to seniors is 3 : 2.
Each student travels to school by Bus, by Car, or Walks.

- 45% of the 400 students take the Bus.
- Of the seniors, 30 take the Car and 50 Walk.
- There are 80 juniors who take the Car.

(a) Complete the two-way table to classify this data.

	Bus	**Car**	**Walk**	**Total**
Juniors				
Seniors				
Total				400

[3]

(b) A student is chosen at random.
Find the probability that they are a junior who walks.

..... **[1]**

(c) Two seniors are chosen at random.
Find the probability that they both take the bus.

..... **[2]**

7 80 students travel to school either by Bus or by Walking.
The two-way table shows the number of boys and girls using each method of transport, in terms of x.

	Bus	**Walk**	**Total**
Boys	x	$2x$	
Girls	$x + 5$	$3x - 2$	
Total			80

(a) Work out the value of x.

$x =$ ..... **[3]**

(b) Complete the two-way table with the number of students in each category.

	Bus	**Walk**	**Total**
Boys			
Girls			
Total			80

[2]

(c) A student is chosen at random from those who walk to school.
Find the probability that this student is a girl.

..... **[2]**

8 A company has 300 employees. Each employee is either a Manager, an Office Worker, or a Technician.
Each employee works either Full-Time or Part-Time.

- The ratio of Managers : Office Workers : Technicians is 2 : 5 : 3.
- 20% of the Managers work Part-Time.
- 40% of the Office Workers work Part-Time.
- The number of Part-Time Technicians is equal to the number of Part-Time Managers.

(a) Complete the two-way table for this information.

	Full-Time	**Part-Time**	**Total**
Managers			
Office Workers			
Technicians			
Total			

[4]

(b) A Part-Time employee is chosen at random.
Find the probability that they are an Office Worker.

..... **[2]**

(c) Two Managers are chosen at random.
Find the probability that exactly one of them works Part-Time.

..... **[3]**

9 A group of 120 students are classified by gender (Boys, Girls) and handedness (Left-handed, Right-handed).
The two-way table shows the number of students in each category in terms of x and y.
Two students are chosen at random from the 120 students.
The probability that both students are left-handed is $\frac{3}{34}$.

	Left-handed	**Right-handed**
Boys	x	y
Girls	$2x$	$y - 6$

(a) Show that $3x^2 - x - 420 = 0$.

[4]

(b) Find the value of x and the value of y.

$x =$ ..... , $y =$ ..... **[4]**

(c) A right-handed student is chosen at random.
Find the probability that the student is a girl.

..... **[2]**